

TradePilot v5 -- Master Research Document

Definitive reference for building TradePilot v5: multi-horizon, ML-powered, India-first

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1. Executive Summary

TradePilot v5 is a complete architecture rebuild: from single-horizon classification to **multi-horizon ensemble ML** with regime-aware risk management. The core thesis remains unchanged -- build the platform that makes Indian traders profitable -- but the engine underneath gets radically smarter.

Key numbers driving v5:

Metric	Value	Source
Indian demat accounts	212.8M	CDSL/NSDL 2025
F&O traders losing money	93%	SEBI 2024
Aggregate retail F&O losses	Rs 1.8 trillion (3yr)	SEBI 2024
Monthly SIP flows (structural floor)	Rs 29,361 cr	AMFI Sep 2025
Backtest Sharpe degradation in live	30-50%	Academic consensus

Metric	Value	Source
LightGBM vs deep learning on tabular	LightGBM wins	Kaggle + papers
Insider buying cluster outperformance	11.2% over 6 months	SEBI data analysis
Target portfolio Sharpe (live)	>1.0	v5 design goal

What v5 delivers: 4 trading pools (intraday/swing/positional/investment), LightGBM+TFT+LSTM ensemble, regime detection via HMM, alternative data (FII flows, insider trades, delivery %), automated circuit breakers, and tax-aware profit waterfall -- all running on an M1 Mac.

| 2. V4 to V5 Evolution

What Failed in V4

V4 Problem	Impact	V5 Fix
Classification target (up/down/avoid)	Destroyed magnitude info; 96% AVOID class	Regression target: predict intraday return magnitude
Single horizon (intraday only)	Missed swing/positional alpha	4 independent pools with separate models
No regime detection	Same strategy in bull and bear	HMM 3-state model shifts allocation dynamically
k-fold cross-validation	Data leakage from future	Walk-forward with 5-day embargo
No alternative data	Only price/volume features	FII/DII flows, delivery %, insider buys, VIX
No risk framework	Position sizing by gut feel	VaR/CVaR + Kelly + ATR stops per horizon
No compounding rules	Profits sat idle	Profit waterfall across pools monthly

What Stays from V4

- LightGBM as primary model (validated as best for tabular financial data)
- Core feature set: RSI, VWAP deviation, ATR, volume ratio, gap %
- Zerodha/Dhan broker integration
- pandas-ta for feature engineering
- vectorbt for backtesting research

| 3. Multi-Horizon Architecture

3.1 Four Pool Structure (Rs 50L base capital)

Pool	Allocation	Capital	Holding Period	Turnover	Model
Intraday	30%	15L	Close by 3:15 PM	Daily	LightGBM (30-min return)
Swing	25%	12.5L	3-7 days	Weekly	LightGBM + TFT ensemble
Positional	25%	12.5L	2-4 weeks	Bi-weekly	Sector rotation + momentum
Investment	15%	7.5L	1-6 months	Monthly	Fundamental + insider signals

Pool	Allocation	Capital	Holding Period	Turnover	Model
Reserve	5%	2.5L	Drawdown buffer	On-demand	Cash only

3.2 Regime-Based Allocation Shifts

Composite regime score from 6 indicators (Nifty vs 50/200-DMA, VIX, A/D ratio, FII 5d flow, sector breadth). Bull $\geq +3$, Bear ≤ -3 , else Sideways.

Pool	Bull	Sideways	Bear
Intraday	30%	35%	25%
Swing	30%	20%	15%
Positional	25%	20%	10%
Investment	15%	15%	20%
Reserve	0%	10%	30%

3.3 Profit Waterfall (Monthly Cycle)

Intraday profits -> 50% stays, 30% -> swing, 20% -> positional
Swing profits -> 60% stays, 40% -> investment
Positional profits -> 70% stays, 30% -> investment
Investment -> 100% reinvested (true compounding)

3.4 Rebalancing Rules

- Weekly drift check: rebalance if any pool > 5% off target
- After drawdown: reduce pool to 75% for 2 weeks, excess to reserve
- Restore full size only after 5 consecutive profitable days

| 4. Data Sources Master Table

Ranked by predictive power for Indian markets. All free unless noted.

4.1 Tier 1 -- Highest Alpha (Always Include)

Source	Data Type	Alpha Signal	Frequency	Python Library
NSE FII/DII flows	Cash + F&O activity	FII selling >2000cr = bearish 1-3 days	Daily EOD	<code>nsefin</code>
Participant-wise OI	FII/DII/Pro/Client positions	FII long/short ratio predicts direction	Daily 5-6 PM	<code>nsepython</code>
NSE delivery bhavcopy	Delivery % per stock	>60% + price up = institutional buying	Daily EOD	<code>NseIndiaApi</code>
India VIX	Fear gauge	<11 complacency, >25 panic buy signal	Real-time	<code>jugaad-data</code>
GIFT Nifty	Overnight price discovery	Gap prediction ~75% accuracy	Continuous	TradingView API
Insider trades (SEBI)	Promoter buys/sells	3+ insiders in 30d = 11.2% outperformance	Weekly	SEBI SAST filings

4.2 Tier 2 -- Strong Signal (Include After Validation)

Source	Data Type	Alpha Signal	Frequency	Access
NSE option chain	OI + Greeks by strike	Max pain convergence on expiry day	Real-time	<code>nsepython</code>
Promoter pledge data	Pledged shares %	Rising pledges = risk of forced selling	Quarterly	BSE/NSE filings
Bulk/block deals	Large transactions	Institutional entry/exit signals	Daily	NSE website
AMFI MF flows	Category-wise inflows	SIP decline 3mo = market top signal	Monthly	AMFI website
Nifty sectoral indices	Sector performance	Relative strength for rotation model	Daily	NSE website
Google Trends (pytrends)	Search interest	Retail sentiment proxy	Weekly	<code>pytrends</code>

4.3 Tier 3 -- Alternative / Supplementary

Source	Data Type	Alpha Signal	Frequency	Access
UPI transaction volumes	NPCI monthly data	Consumer spending proxy	Monthly	NPCI website
Monthly auto sales	SIAM/FADA data	Sector health indicator	Monthly	Industry reports
Naukri JobSpeak Index	Hiring activity	Economic momentum proxy	Monthly	Naukri website
FinBERT sentiment	News sentiment scoring	Contrarian at extremes	Real-time	HuggingFace (free)
MSCI rebalancing	Index additions/deletions	Forced FII flows \$200-500M	Quarterly	MSCI website

4.4 Paid (Worth It)

Source	Cost	What You Get
Zerodha Kite Connect full	Rs 500/mo	10yr intraday data + WebSocket 3000 instruments
Chartink alerts	Rs 780/mo	Webhook-triggered scans, 100+ indicators
Trendlyne Pro	Rs 119/mo	1400+ parameters, DVM scoring

| 5. ML Engine Design

5.1 Ensemble Architecture

```
Input Features (20)
|
+---> LightGBM Regressor -----> predicted_return_lgb (weight: 0.50)
|   (tabular, fast, interpretable)
|
+---> Temporal Fusion Transformer -> predicted_return_tft (weight: 0.30)
|   (attention over time, handles regime shifts)
|
+---> LSTM -----> predicted_return_lstm (weight: 0.20)
|   (sequential patterns, momentum)
|
v
Weighted Average -> final_predicted_return
|
v
RL Position Sizer (FinRL PPO) -> position_size (% of pool)
|
v
Risk Filter (VaR, regime, circuit breaker) -> TRADE or SKIP
```

Expected live performance: Sharpe 1.0-1.8, IC 0.05-0.10 (30-50% below backtest).

5.2 Target Variable

```
# Regression, NOT classification (v4's mistake)
target = (close_1500 - open_0930) / open_0930
target = np.clip(target, target.quantile(0.01), target.quantile(0.99)) # winsorize
```

5.3 Top 20 Features (SHAP-ranked)

Tier 1 (always include): RSI(14), VWAP deviation, ORB 15-min range, previous day return, ATR(14)/close, volume ratio (today/20d avg), gap %.

Tier 2 (include after validation): MACD histogram, Bollinger %B, ADX(14), FII/DII net flow, options OI PCR, sector relative strength, 5-day return, intraday volatility (H-L)/C.

Tier 3 (calendar/alternative): Day of week, days to F&O expiry, Nifty 50 morning return, India VIX, delivery %.

5.4 LightGBM Configuration

```
LGBM_PARAMS = {
    'objective': 'regression', 'metric': 'mae',
    'num_leaves': 31, 'max_depth': 6,
    'min_child_samples': 100, # HIGH -- prevents noise fitting
    'reg_alpha': 0.1, 'reg_lambda': 1.0,
    'subsample': 0.7, 'colsample_bytree': 0.7,
    'learning_rate': 0.01, 'n_estimators': 5000, # early stopping cuts
}
```

5.5 Validation: Walk-Forward with Embargo

```
Fold 1: [====train 1yr====]--5d embargo--[test 1mo]
Fold 2:  [====train 1yr====]--5d embargo--[test 1mo]
Fold 3:  [====train 1yr====]--5d embargo--[test 1mo]
```

Success criteria: IC > 0.05 in >60% of folds. Long-short spread > 0 in >70% of months after costs.

5.6 Training Pipeline

Step	Tool	Frequency
Feature computation	pandas-ta + custom	Daily batch
Feature store	DuckDB (Parquet files)	Append daily
Model training	LightGBM + neuralforecast (TFT/LSTM)	Monthly retrain
Hyperparameter search	Optuna	Quarterly
Feature importance	SHAP + alphasens-reloaded	Monthly
Backtest validation	vectorbt	After every retrain
Signal generation	Ensemble inference	Real-time
Position sizing	FinRL PPO	Trained quarterly

5.7 Foundation Models (Supplementary Only)

TimesFM (Google) and Chronos (Amazon) are mediocre alone but add 2-5% ensemble improvement. Use as a 4th ensemble member with weight 0.10 if compute allows. All run on M1 Mac.

6. Indian Market Edges -- Top 10 Ranked

Ranked by (alpha potential x accessibility). All free data.

Rank	Edge	Alpha	Effort	Horizon	Priority
1	India VIX regime detection	High	Low	All	P0
2	FII/DII flow tracking	High	Medium	Swing + Positional	P0
3	GIFT Nifty gap prediction	Medium	Low	Intraday	P0
4	Delivery % signals	Medium	Medium	Swing	P1
5	Insider buying clusters	High	Medium	Positional + Investment	P1
6	F&O expiry patterns	Medium	Medium	Intraday + Swing	P1
7	Sector rotation model	High	High	Positional + Investment	P1
8	Promoter pledge stress	Medium	Low	Investment	P2
9	MF flow analysis	Low-Med	Low	Investment	P2
10	Small/mid-cap momentum ratio	High (risky)	Medium	Positional	P2

Key Signals Summary

- **VIX < 11:** Buy puts for protection. **VIX > 25:** Aggressive equity buying.
- **FII net sell > 2000cr/day:** Bearish 1-3 sessions. **FII buy reversal after 10+ sell days:** Strong rally.
- **GIFT Nifty premium > 0.3%:** Gap-up 75% probability. Trade at open.
- **Delivery > 60% + price up 3%:** Institutional accumulation. Buy for swing.
- **3+ insiders buying in 30 days:** 11.2% outperformance over 6 months.
- **Weekly expiry now Tuesday (NSE, since Sep 2025).** All expiry models must recalibrate.

7. Alternative Data Integration

What V5 Adds Beyond Price/Volume

Data Category	Specific Source	Integration Method	Model Input
Institutional flows	FII/DII daily CSV from NSE	<code>nsefin</code> Python lib, daily batch	Feature: fii_net_5d, dii_net_5d

Data Category	Specific Source	Integration Method	Model Input
Options market	Participant-wise OI, option chain	<code>nsepython</code> , daily batch	Feature: fii_long_short_ratio, pcr
Delivery conviction	NSE delivery bhavcopy	<code>NseIndiaApi</code> , daily batch	Feature: delivery_pct, delivery_vs_20d_avg
Insider activity	SEBI SAST filings, bulk/block deals	Web scrape weekly	Feature: insider_buy_cluster_30d
Sentiment	FinBERT on news headlines	HuggingFace inference, hourly	Feature: news_sentiment_score
Search interest	Google Trends for stock/sector names	<code>pytrends</code> , weekly	Feature: search_momentum_7d
Macro proxy	UPI volumes (NPCI), auto sales	Monthly manual + DB	Feature: consumer_momentum_index
Volatility regime	India VIX + GARCH(1,1)	<code>arch</code> library, daily	Feature: vix_regime, garch_vol

India-Specific Notes

- No credit card transaction data, dark pools, or satellite imagery alpha in India
 - **Uniquely Indian edges:** Promoter pledging (no Western equivalent), bulk/block deal transparency, daily FII/DII disclosure (most markets don't publish daily)
-

| 8. Risk Management Framework

8.1 Drawdown Limits

Horizon	Max Daily DD	Max Weekly DD	Max Monthly DD	Kill Switch
Intraday	2% of pool	5% of pool	10% of pool	Pause 1 day
Swing	--	3% of pool	8% of pool	Reduce size 50%
Positional	--	--	10% of pool	Exit weakest 50%
Investment	--	--	15% of pool	Review, no panic
Portfolio	1% of total	3% of total	7% of total	All-stop

8.2 Circuit Breaker Rules

Trigger	Action	Duration
Pool daily DD > 2%	Pause pool rest of day	1 day
Pool weekly DD > 5%	Reduce pool to 50% size	1 week
Portfolio weekly DD > 3%	Pause all intraday/swing	2 days
Portfolio monthly DD > 7%	All-stop, reserve only	Until manual review
5 consecutive losing days	Reduce all pools to 50%	3 days
Single trade > 1% total capital loss	Review sizing model	Immediate

8.3 Recovery + Regime Detection

Recovery ladder: After circuit breaker: Day 1-3 at 25%, Day 4-7 at 50%, Day 8-14 at 75%, Day 15+ full.

Regime detection: 3-state HMM on (returns, volatility) via `hmmlearn`. Current state drives allocation (Section 3.2) + VIX sizing: `size_multiplier = min(15/current_vix, 1.0)`.

8.5 Stop-Loss by Horizon

Horizon	Method	Typical %	Trailing
Intraday	ATR(14) x 1.5	1.5-2%	1 ATR after 1:1 R:R
Swing	ATR(14) x 2.5 + support	3-5%	Close below 8-EMA
Positional	Weekly ATR x 2 + 200-DMA	5-8%	Close below 21-EMA

Horizon	Method	Typical %	Trailing
Investment	85% of fair value estimate	10-15%	None; quarterly review

8.6 Correlation Guard

Never hold >3 stocks from same sector across all pools. If correlation > 0.7 between two holdings, treat as 1.5 effective positions for Kelly sizing.

9. Technology Stack

9.1 Core Stack

Layer	Tool	Why
Data ingestion	Dhan API (free, dev) + Zerodha Kite (prod)	Zero cost dev, best liquidity prod
Historical data	jugaad-data + OpenChart (1-min)	Free, 10+ years daily, 5yr intraday
Tick storage	QuestDB	ASOF JOIN (critical for financial data), fastest ingestion
Feature store	DuckDB + Parquet	Zero infra cost, fast analytics, AES-256
Backtesting	vectorbt (research sweeps) + Backtrader (event-driven)	Speed + live trading path
Feature engineering	pandas-ta (primary) + TA-Lib (optional C accelerator)	150+ indicators, pure Python
ML training	LightGBM + neuralforecast (TFT, LSTM)	Best tabular + best temporal
RL sizing	FinRL (PPO agent)	Position sizing > signal generation for RL
Hyperparameters	Optuna	Bayesian optimization, pruning
Feature analysis	SHAP + alphasens-reloaded	Importance + factor tearsheets
Research platform	Microsoft Qlib (15K stars)	End-to-end quant research
Portfolio optimization	PyPortfolioOpt (quick) + Riskfolio-Lib (advanced)	MV, HRP, CVaR, risk parity
Execution abstraction	OpenAlgo	30+ Indian brokers, one unified API
Screening	Chartink webhooks + custom scanner	Free, webhook to OpenAlgo

Layer	Tool	Why
Dashboards	Grafana + QuestDB	Real-time P&L, portfolio heat maps
Alerts	Telegram Bot API	What Indian traders actually use
Sentiment	FinBERT / FinGPT (HuggingFace)	Free, runs on M1 Mac
Performance reporting	quantstats	Full tearsheets, benchmark comparison

9.2 Broker API Ranking (India)

Rank	Broker	Latency	Cost	Best For
1	Dhan	~50ms	Free	Development, intraday algo
2	Zerodha Kite	Low	Rs 500/mo	Production, largest ecosystem
3	Fyers	Low	Free	Free API with bracket orders
4	Angel One SmartAPI	Moderate	Free	Easiest onboarding
5	Flattrade	Moderate	Free + 0 brokerage	Cost minimization

9.3 Key Dependencies

All run on M1 Mac, no GPU required. Core: `lightgbm neuralforecast vectorbt pandas-ta pyportfolioopt riskfolio-lib hmmlearn shap alphalens-reloaded optuna quantstats nsefin jugaad-data nsepython finrl questdb duckdb python-telegram-bot`.

| 10. Implementation Roadmap

Phase 1 -- Foundation (Weeks 1-3): QuestDB + DuckDB setup. Ingest 5yr daily (jugaad-data) + 1yr 1-min (OpenChart). Build 20-feature pipeline (pandas-ta + custom). Integrate FII/DII daily feed via `nsefin`. Set up walk-forward validation with 5-day embargo.

Phase 2 -- ML Engine (Weeks 4-6): Train LightGBM regressor (IC > 0.05 target). Train TFT + LSTM via neuralforecast. Build ensemble combiner (weighted average + confidence). SHAP analysis to trim to 12-15 features. Full backtest via vectorbt.

Phase 3 -- Risk + Execution (Weeks 7-9): HMM regime detector (3-state). 4-pool capital manager with rebalancing. Circuit breakers + ATR stop engine. OpenAlgo integration for paper trading. Dhan API as dev broker.

Phase 4 -- Alt Data + Production (Weeks 10-12): Insider trade scraper (SEBI filings). Delivery % + FinBERT sentiment features. Retrain ensemble with alt data. Grafana dashboards + Telegram alerts. Tax-aware P&L tracker. Zerodha production broker switch.

Total: 12 weeks to live trading.

| 11. Success Criteria

11.1 ML Model Metrics

Metric	Target	Measurement
Information Coefficient (IC)	>0.05 mean	Walk-forward folds
IC stability	Positive in >60% of folds	Walk-forward folds
Long-short spread	>0 in >70% of months	After costs (0.1%/trade)
Hit rate (intraday)	>53%	Live trades
Hit rate (swing)	>50%	Live trades

11.2 Portfolio Metrics

Metric	Target	Horizon
Live Sharpe ratio	>1.0	Annual
Sortino ratio	>2.0	Annual
Calmar ratio	>2.0	Annual

Metric	Target	Horizon
Max drawdown per pool	<10%	Any period
Max portfolio drawdown	<7%	Any period
Profit factor	>1.5	Per pool
Win rate (overall)	>55%	All trades
Annual return target	25-40%	After costs + taxes

11.3 Operational Metrics

Metric	Target
Model retrain frequency	Monthly
Feature store lag	<1 hour from market close
Signal generation latency	<5 seconds
Order execution (Dhan/Zerodha)	<100ms
Dashboard refresh	Real-time (WebSocket)
Uptime during market hours	99.9%

| 12. Key Research References

Open-Source Repositories

Repository	Stars	Use In v5
machine-learning-for-trading (Stefan Jansen)	17,009	ML cookbook, LightGBM patterns
QuantConnect Lean	18,288	Strategy reference
FinRL	14,698	PPO position sizing agent
Microsoft Qlib	15,000+	End-to-end quant research platform
vectorbt	7,107	Fast backtesting
OpenAlgo	1,608	30+ Indian broker abstraction
jugaad-data	504	Free NSE historical data

Proven Strategies (Backtested)

Strategy	Sharpe	Max DD	Best Horizon
VWAP mean reversion	3.57	16%	Intraday
Opening Range Breakout (ORB)	2.40	--	Intraday
Momentum factor (top decile)	--	--	Positional
Pairs trading (Nifty 50 cointegrated)	--	2.57%	Market-neutral
LightGBM 30-min direction	1.0-1.8 (live)	--	Intraday + Swing

Academic / Industry Sources

- SEBI: "Analysis of Profit and Loss of Individual Traders in Equity F&O Segment" (Sep 2024)
- Lopez de Prado: "Advances in Financial Machine Learning" (mlfinlab implementation)
- Stefan Jansen: "Machine Learning for Algorithmic Trading" (2nd edition)
- QuestDB benchmarks: 12-36x faster than InfluxDB for ASOF JOIN
- Kaggle competitions: LightGBM consistently outperforms deep learning on tabular data

Research compiled: April 8, 2026. All tools verified, all data sources confirmed free/accessible. This document is the single source of truth for TradePilot v5 implementation.